Ramp energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare gravitational energy change with measured kinetic energy.

SUCCESS CRITERIA - I can...

1. Measure height vertically and speed at a stated location.
2. Calculate an energy balance.
3. Discuss rotation, resistance and uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “rotation” and “thermal energy” to compare your model or result. Sentence starters are optional.

rotation: turning about an axis

thermal energy: internal energy associated with microscopic motion/interactions

energy budget: an account of all energy stored and transferred

01 Warm-up: bring back an earlier idea

An ideal system has total mechanical energy 15 J. At a point where U=3 J, find K.

Word help: mechanical energy = total kinetic and potential energy in a model

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: mg Δh = $\Delta K_{translation}$ + $\Delta K_{rotation}$ + $\Delta E_{thermal}$. Closed energy accounting near Earth; Δh is the positive drop. Determine whether the object rolls or slides; use the approved ramp setup.

Worked reasoning: A supplied ramp run loses 6 J of gravitational potential energy but gains only 4 J of translational kinetic energy. The 2 J difference needs an account, such as rotation or friction.

Your turn: Draw an energy account with the 2 J remainder explicitly labeled as not yet identified.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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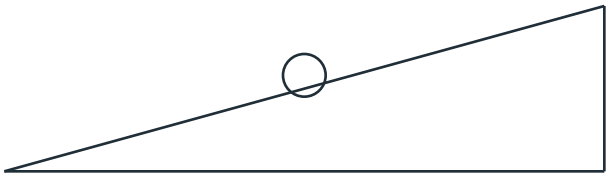
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

On the ramp sketch, mark the vertical height h and the along-ramp path. Which belongs in mgh?

Sentence starter: The ... in my model represents ... . Its label shows ... .



Label h vertically. The marker represents the object on the sloping surface.

Marzano check - circle: 1 2 3

04 Apply the physics

Supplied model: mass 1 kg drops 0.50 m; g=10. At bottom translational K=4 J.
Find lost U and energy not accounted for by translation alone.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Create a table for repeated trials. Include actual height, mass, bottom speed,ΔU, K and one observation about losses or rotation.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Keep mass fixed and multiply speed by 4. By what factor does kinetic energy change?
Explain from the equation.
Show the energy changes and transfers; label units.

Word help: kinetic energy = energy associated with motion

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A rolling object has less½mv² than its lost mgh.
Name two possible destinations of the difference.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Design a comparison that helps distinguish rolling energy from sliding friction as explanations for a lower translational speed.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3